

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

11

Q1

GRID-IN ALGEBRA

The equation $7g + 7b = 840$ represents the number of blue tiles, b , and the number of green tiles, g , an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A line segment that has a length of 115 centimeters (cm) is divided into three parts. One part is 47 cm long. The other two parts have lengths that are equal to each other. What is the length, in cm, of one of the other two parts of equal length?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ALGEBRA

The relationship between two variables, x and y , is linear. For every increase in the value of x by 1, the value of y increases by 8. When the value of x is 2, the value of y is 18. Which equation represents this relationship?

- A. $y = 2x + 18$
- B. $y = 2x + 8$
- C. $y = 8x + 2$
- D. $y = 3x + 26$

Q4

ALGEBRA

$$x + 2y = 6$$

$$x - 2y = 4$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. 2.5
- B. 5
- C. 6
- D. 10

$$y > 7x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	13
5	27
8	48

B.

x	y
3	17
5	31
8	52

C.

x	y
3	21
5	27
8	52

D.

x	y
3	21
5	35
8	56

Q6

ALGEBRA

Lily made 36 cups of jam. Lily then filled x small containers and y large containers with all the jam she made. The equation $4x + 6y = 36$ represents this situation. Which is the best interpretation of $6y$ in this context?

- A. The number of large containers Lily filled
- B. The number of small containers Lily filled
- C. The total number of cups of jam in the large containers
- D. The total number of cups of jam in the small containers

$$66x = 66x$$

How many solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q8

GRID-IN ALGEBRA

The function f is defined by $fx = 4x + kx - 1$, where k is a constant, and $f5 = 32$.
What is the value of $f10$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$-15x + 25y = 65$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which of the following could be the second equation in the system?

- A. $12x + 20y = 52$
- B. $12x + 20y = -52$
- C. $-12x + 20y = 52$
- D. $-12x + 20y = -52$

$$H = 120p + 60$$

The Karvonen formula above shows the relationship between Alice's target heart rate H , in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice's target training heart rate?

- A. $120 \leq H \leq 162$
- B. $102 \leq H \leq 120$
- C. $60 \leq H \leq 162$
- D. $60 \leq H \leq 102$